

535/1
PHYSICS
Paper 1
2026



Uganda Certificate of Education
PREMOCK 2026
PHYSICS Theory
Paper 1
2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

This paper consists of two sections; A and B.

It has seven examination items.

Section A has three compulsory items.

Section B has two parts; I and II. Answer one item from each part.

Answer five items in all.

Any additional item(s) answered will not be scored.

All answers must be written in the booklets provided.

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SECTION A

Answer all the items from this section

Item 1

A community hall was hosting a music concert. Two students, Akello and Okello, were standing at different positions from the stage. Akello heard the sound 3 seconds after the performance started, while Okello heard it 5 seconds after. During the concert, coloured stage lights (red, green, and blue) were shone on the performers who were wearing white costumes. The audience noticed that the costumes appeared different colours under different lights.

Hint: Speed of sound in air = 330 m/s

Task:

As a physics student, help the students to understand why:

- (a) Akello and Okello heard the sound at different times.
- (b) If the performers were originally wearing white costumes, explain the colour changes when:
 - Red light was shone on them
 - Green light was shone on them
 - Blue light was shone on them
- (c) Why the sound from the speakers in the hall was clear near the stage but became unclear and confused at the back of the hall.

Item 2

A medical facility uses radioactive isotopes for cancer treatment. The hospital administrator is concerned about the safe disposal of radioactive waste. A sample of radioactive material has an initial activity of 1600 counts per minute and a half-life of 8 days. The safety regulations require that radioactive waste must have an activity below 25 counts per minute before disposal.

Task:

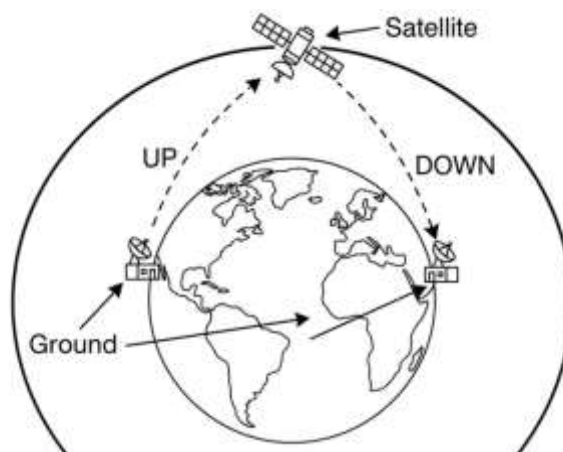
As a student of physics:

- (a) Determine the number of days the hospital must wait before disposing of the radioactive waste safely.
- (b) Sensitise the hospital staff about:
 - The risks associated with handling radioactive materials
 - The safety precautions that must be taken when handling radioactive materials
 - The proper methods of storing radioactive materials

Item 3

A satellite in orbit around the Earth transmits television signals to different countries. People in Uganda are watching a live football match being played in Brazil during the day, while people in Brazil are watching the same match at night. The viewers are confused about how this is possible.

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Task:

Using your knowledge of physics, help the viewers to understand:

- Why it is day in Uganda while it is night in Brazil at the same time.
- The role of satellites in broadcasting live events from one country to another.
- The type of orbit a communication satellite must have and why this type of orbit is suitable for communication.

SECTION B

PART I

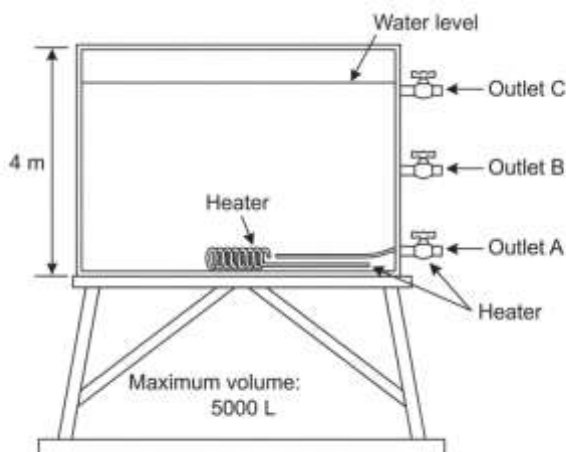
Answer one item from this part

Item 4

A school is installing a solar water heating system on the roof of a two-storey building. The water tank has a capacity of 3000 litres and is placed on a metallic stand. The water inlet pipe is connected to the main water supply at ground level, and the outlet pipe is at the bottom of the tank. The solar heater is fitted at the bottom of the tank.

Data:

- Density of water = 1000 kg/m^3
- Specific heat capacity of water = $4200 \text{ J/kg}\cdot\text{K}$
- Acceleration due to gravity = 10 m/s^2



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Task:

As a student of physics:

(a) Explain to the school administration why:

- The solar heater is fitted at the bottom of the tank.
- The outlet pipe is at the bottom of the tank and not at the top.

(b) If the water in the tank is at 15°C and the solar heater supplies 252,000 kJ of energy, determine the final temperature of the water.

(c) Advise the school on measures that can be taken to ensure that the tank stand can support the weight of the tank and water for a long time.

Item 5

A restaurant on the 5th floor of a building needs hot water for washing dishes. A boiler on the ground floor heats water to 90°C . The restaurant provides 15 litres of cold water at 25°C to each customer for washing. The restaurant manager wants to know how much hot water to mix with the cold water to achieve a washing temperature of 40°C . The manager also wants to keep the water hot without continuously boiling it.

Data:

- Density of water = 1000 kg/m^3
- Specific heat capacity of water = $4200 \text{ J/kg}\cdot\text{K}$
- Acceleration due to gravity = 10 m/s^2

Task:

Having studied physics:

(a) Help the manager determine the quantity of hot water (at 90°C) to be mixed with 15 litres of cold water (at 25°C) to achieve a final temperature of 40°C .

(b) Advise the manager on how to keep the boiled water hot for a long period of time without keeping the boiler on.

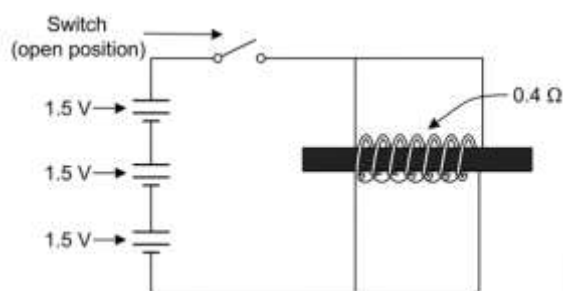
(c) Explain how the water from the boiler on the ground floor can reach the 5th floor safely.

PART II

Answer one item from this part

Item 6

A recycling company needs to separate iron nails from a mixture of non-magnetic materials. The available materials are: an iron rod, copper wire of resistance 0.4Ω , and three dry cells each of 1.5 V.



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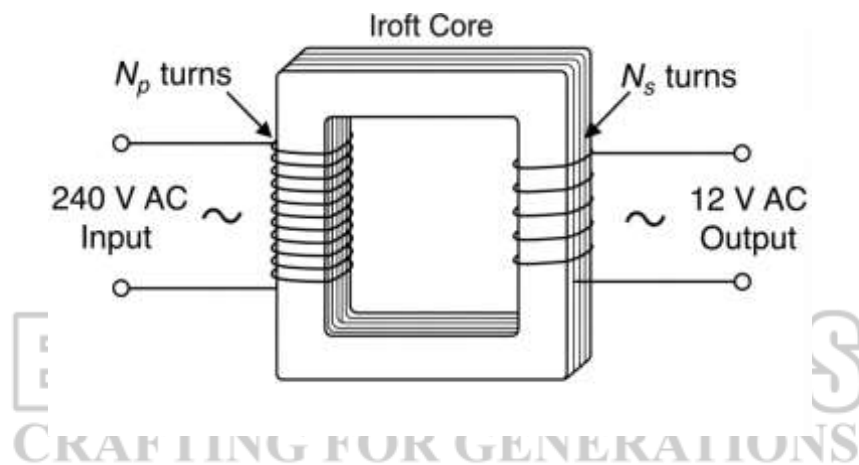
Task:

As a student of physics:

- Design and describe an electrical method the company can use to separate iron nails from the mixture.
- Comment on the effectiveness of your design, given that a current of 5 A is required to create a sufficiently strong magnet for this purpose.
- Explain how the strength of the electromagnet can be increased if needed.

Item 7

A school has been connected to the national grid where electricity is transmitted at 240 V. The school intends to install computers that operate at 12 V DC. The school administrator does not understand how to convert the 240 V AC to 12 V DC.

**Task:**

As a student of physics, help the school administrator to understand:

- What a transformer is and how it works.
- What type of transformer is needed to change 240 V to 12 V AC, and determine the ratio of turns on the primary coil to the secondary coil.
- How the 12 V AC from the transformer can be converted to 12 V DC for the computers.
- Why it is important to have all computers connected in parallel rather than in series.

END

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